

Jackson Yuan

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 Coquitlam, BC

CORE QUALIFICATIONS

- Proficiency with back-end software development languages, such as **C++/C, Java, Python**
- Proficiency with back-end frameworks: **Spring Boot, Spring MVC and RESTful**
- Proficiency with **GitHub, Windows OS, MacOS and Linux(Ubuntu) system**
- Familiar with Query Language, such as **MySQL, Microsoft SQL server**
- Self-learning front-end software development, such as **HTML, CSS, JavaScript**, and framework **React.js, Vue.js**
- Familiar with data analysis, such as **RStudio, SAS**
- Excellent knowledge at **MATLAB, AWS and LaTeX**

EDUCATION

Bachelor of Applied Science - Computer Science

Jan 2021 – Dec 2025

Minor in **Statistics**

Simon Fraser University

Burnaby, BC

TECHNICAL PROJECTS

Smart Ordering System (Full-Stack System)

Dec 2025 - Present

Technologies: Java / Spring Boot / REST API / MySQL / HTML / CSS / JavaScript

- Designed and implemented a full-stack restaurant ordering system including customer ordering, kitchen dispatch, and management workflows
- Built RESTful backend services using Spring Boot to handle order lifecycle, menu management, and status synchronization
- Implemented role-based system architecture separating client, admin, and service layers for scalability
- Designed relational database schemas to support orders, items, users, and transaction records

Multi-Agent Path Finding (MAPF) with Prioritized Planning & CBS

Nov 2025

Technologies: Python / A / MAPF / CBS*

- Implemented Multi-Agent Path Finding (MAPF) solvers using Prioritized Planning and Conflict-Based Search (CBS)
- Designed constraint handling for vertex and edge conflicts across multiple agents
- Compared algorithmic performance in terms of success rate, runtime, and memory usage
- Demonstrated significant performance improvements of CBS over baseline approaches in complex environments

Statistical Learning and Predictive Modeling

July 2024

R / Random Forest / Boosting / LASSO / Ridge / SVM / ROC-AUC

- Developed and compared multiple machine learning models for regression and classification, including Random Forest, Boosting, LASSO, Ridge, LDA, and SVM.
- Evaluated models using MSPE, ROC/AUC, and classification error, with hyperparameter tuning for performance optimization.

Weather APP (Web)

Apr 2023

- Used **HTML and CSS** to construct the structure of Website
- Gave a format and structure for the future development whole index website
- Used **Open Weather API** to query all information